

2019

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Remove the multiple-choice answer sheet at the back of Section I
- NESA approved calculators may be used
- A formula sheet, data sheet and Periodic Table are provided at the back of this paper
- For questions in Section II, show all relevant working in questions involving calculations

Total marks:
100

Section I — 20 marks (pages 2-8)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II — 80 marks (pages 9-25)

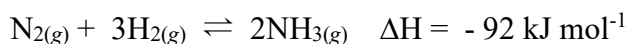
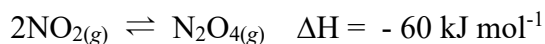
- Attempt Questions 21–39
- Allow about 2 hour and 25 minutes for this section

Section I**20 marks****Attempt Questions 1–20****Allow about 35 minutes for this section**Use the multiple-choice answer sheet for Questions 1–20

1 Which equation would represent the greatest decrease in entropy?

- A. $2\text{O}_{3(g)} \rightarrow 3\text{O}_{2(g)}$
- B. $\text{C}_{(s)} + \text{O}_{2(g)} \rightarrow \text{CO}_{2(g)}$
- C. $2\text{H}_2\text{O}_{2(s)} \rightarrow 2\text{H}_2\text{O}_{(l)} + \text{O}_{2(g)}$
- D. $\text{HCl}_{(g)} + \text{NH}_{3(g)} \rightarrow \text{NH}_4\text{Cl}_{(s)}$

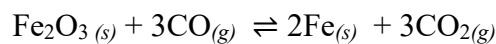
2 Consider the two reactions at equilibrium shown below:



Which action will drive BOTH reactions to form more product?

- A. Reducing the overall pressure
 - B. Increasing the reaction temperature
 - C. Increasing the volume of the reaction vessel
 - D. Removing the product from the reaction vessel
- 3** Under what reaction conditions will the value of an equilibrium constant (K_{eq}) be changed?
- A. Addition of a catalyst
 - B. Increased temperature
 - C. Increased concentration of reactants
 - D. Increased concentration of reactants and products

- 4 Which alternative most correctly represents the equilibrium expression for the following reaction?



- A. $\frac{[\text{CO}_2]^3}{[\text{CO}]^3}$
- B. $\frac{[\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3] [\text{CO}]^3}$
- C. $\frac{[\text{Fe}_2\text{O}_3] [\text{CO}]^3}{[\text{Fe}]^2 [\text{CO}]^3}$
- D. $\frac{[\text{Fe}]^2 [\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3] [\text{CO}]^3}$

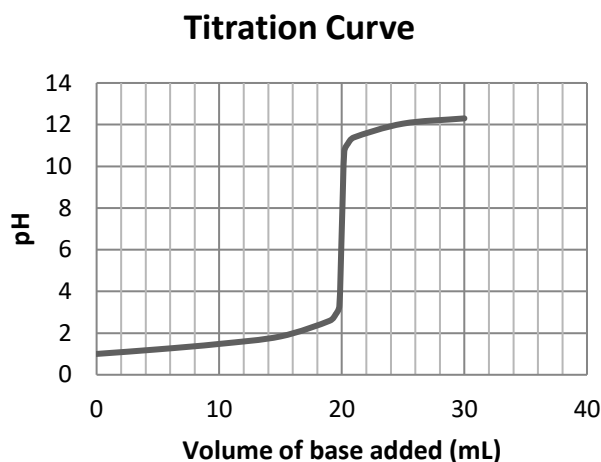
- 5 Which of the following reactions at equilibrium is closest to completion?

- A. $2\text{O}_3(g) \rightleftharpoons 3\text{O}_2(g) \quad K_{eq} = 2.0 \times 10^{57}$
- B. $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g) \quad K_{eq} = 4.3$
- C. $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g) \quad K_{eq} = 9.0$
- D. $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq) \quad K_{eq} = 1.8 \times 10^{-5}$

- 6 Which of the following is a Brønsted–Lowry reaction?

- A. $\text{CO}_2 + \text{OH}^- \rightleftharpoons \text{HCO}_3^-$
- B. $\text{NH}_4^+ + \text{NH}_2^- \rightleftharpoons 2\text{NH}_3$
- C. $\text{HClO}_4 + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{ClO}_4^-$
- D. $\text{CH}_3\text{CH}_2\text{O}^- + \text{CH}_3\text{NH}_3^+ \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{NH}_2$

Use the following information for Questions 7 and 8.



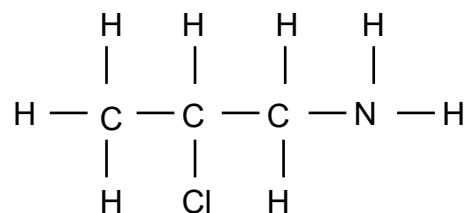
- 7 Which statement about this titration is correct?
- The acid was placed in the burette and the equivalence point was at pH 7.
 - The base was placed in the burette and the equivalence point was at pH 12.
 - The acid was placed in the conical flask and the equivalence point was at pH 7.
 - The acid was placed in the conical flask and the equivalence point was at pH 12.
- 8 Which pair of solutions would be most likely to produce this titration curve?
- Acetic acid and ammonia solution
 - Citric acid and sodium hydroxide
 - Sulfuric acid and ammonia solution
 - Hydrochloric acid and sodium hydroxide
- 9 What is the concentration of H_3O^+ in a solution with a pH of 2.38?
- 0.38 mol L^{-1}
 - 11.62 mol L^{-1}
 - $4.17 \times 10^{-3} \text{ mol L}^{-1}$
 - $2.40 \times 10^{-12} \text{ mol L}^{-1}$

- 10 Which alternative contains only amphoteric substances?
- A. SiO_2 , SO_2 , SnO
 - B. H_2CO_3 , HCO_3^- , CO_3^{2-}
 - C. HS^- , H_3PO_4 , HPO_4^{2-}
 - D. HCO_3^- , HSO_4^- , H_2O
- 11 25.00 mL of 0.02 M potassium hydroxide solution in a conical flask is neutralised by the gradual addition of 33.20 mL of nitric acid.
- What is the concentration of the nitric acid?
- A. 0.015 mol L^{-1}
 - B. $0.0265 \text{ mol L}^{-1}$
 - C. $5 \times 10^{-4} \text{ mol L}^{-1}$
 - D. $1.66 \times 10^{-5} \text{ mol L}^{-1}$
- 12 The formula of potassium hydrogen iodate is:
- A. $\text{HK}(\text{IO}_3)_2$
 - B. $\text{KH}(\text{IO}_3)_2$
 - C. $\text{H}_2\text{K}(\text{IO}_3)_2$
 - D. $\text{K}_2\text{H}(\text{IO}_3)_2$
- 13 Which of the following is the conjugate base of the dihydrogen phosphate ion (H_2PO_4^-)?
- A. OH^-
 - B. PO_4^{3-}
 - C. H_3PO_4
 - D. HPO_4^{2-}

14 Which condensed formula represents a chain isomer for $C_4H_8F_2$?

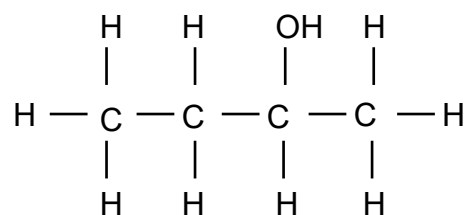
- A. $(CH_3)_2CHFCHF$
- B. $CH_3CHFCHFCH_3$
- C. $CH_3CH_2CHFCHF$
- D. $CH_2FCHCHFCH_3$

15 Consider the following structure.



What is the IUPAC name for this molecule?

- A. 2-chloropropanamide
 - B. 2-chloropropan-1-amine
 - C. 2-chloropropan-3-amine
 - D. 3-amino-2-chloropropane
- 16 Acidified potassium permanganate solution is a powerful oxidising agent. It is added to a substance with the following structural formula;



What is the name of the organic product of this reaction?

- A. Butan-2-ol
- B. Butanal
- C. Butan-2-one
- D. Butanoic acid

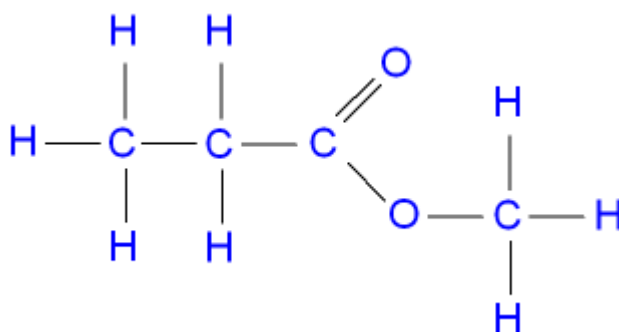
- 17 Three separate colourless solutions each contain one cation, Na^+ , Pb^{2+} or Ca^{2+} . Which of the following would be an appropriate reagent to unambiguously identify the solution containing Pb^{2+} ?
- KI
 - K_2CO_3
 - K_3PO_4
 - AgNO_3
- 18 A colourless organic liquid compound is being tested for the presence of functional groups. 2 mL of the liquid is poured into three separate test tubes. A reagent is added to each test tube then shaken.

<i>Test</i>	<i>Reagent added</i>	<i>Observation made after shaking</i>
1	2 drops of bromine water	Yellowish-brown coloured solution
2	3 drops of acidified potassium dichromate	Yellowish-orange coloured solution
3	1 g of sodium hydrogen carbonate	Colourless solution, small bubbles of colourless gas

What functional group(s) may be present in this compound?

- Double bonds
 - Carboxylic acid
 - Double bonds and hydroxyl group
 - Hydroxyl group bonded to a secondary carbon
- 19 Which of the following is a correct statement about primary and tertiary alcohols?
- Primary alcohols can be oxidised to aldehydes while tertiary alcohols cannot be oxidised.
 - Primary alcohols can be oxidised to aldehydes while tertiary alcohols are oxidised to ketones.
 - Primary alcohols have one hydroxyl functional group while tertiary alcohols have three hydroxyl functional groups.
 - Primary alcohols always have a hydroxyl functional group on the terminal carbon of the parent chain while tertiary alcohols have their functional group on the third carbon of the parent chain.

- 20 The formula of an ester is shown below.



Which alkanol and alkanoic acid were used to make this ester?

- A. Propanol and methanoic acid
- B. Methanol and ethanoic acid
- C. Methanol and propanoic acid
- D. Ethanol and methanoic acid

2019

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Student ID: _____

Chemistry

Section II

Answer Booklet

80 marks

Attempt Questions 21–39

Allow about 2 hours and 25 minutes for this section

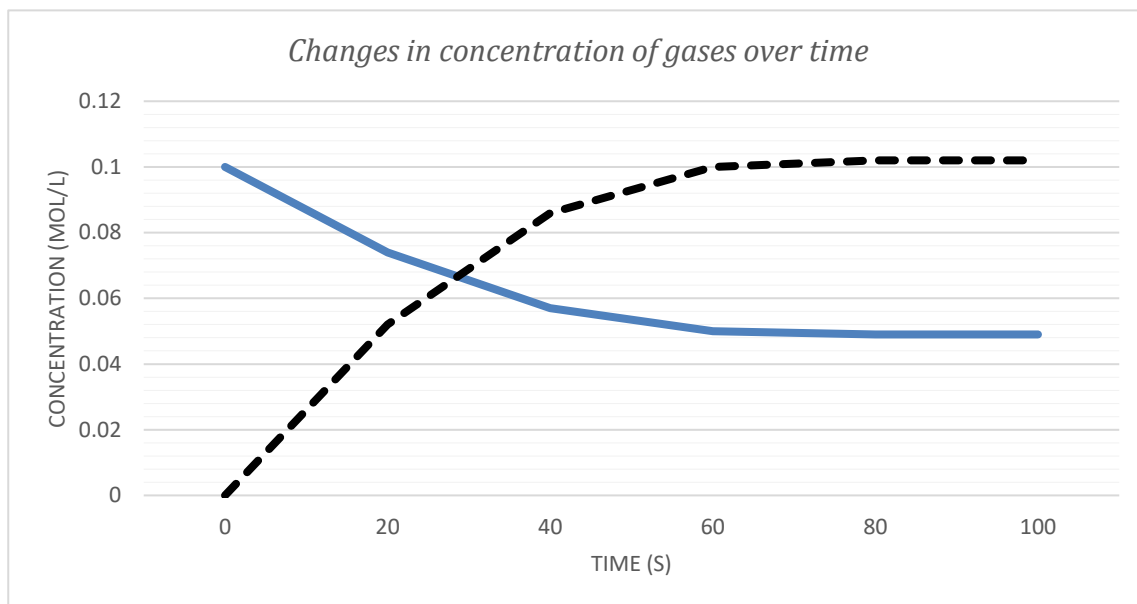
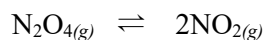
Instructions

- Write your Student ID above
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in question involving calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

Question 21 (4 marks)

A sealed glass flask originally contained colourless dinitrogen tetroxide gas with a concentration of 0.100 mol L^{-1} at a temperature of 100°C . The dinitrogen tetroxide decomposed to form a dark brown gas.



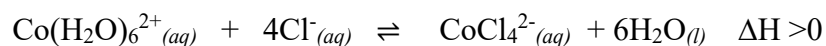
- (a) Describe the macroscopic appearance of the contents of the flask after 80 seconds. 1

- (b) Describe what happened at the molecular level to the contents of the flask after 80 seconds. 1

- (c) Explain the changes in reaction rate from 0 to 80 seconds. 2

Question 22 (4 marks)

A student was provided with a beaker containing a violet coloured solution of cobalt (II) chloride and was asked to investigate the following equilibrium reaction:



- (a) 5 mL of cobalt (II) chloride solution was poured into a test tube with 5 drops of 4 M HCl. **2**

Use Le Chatelier's principle to predict the impact on this reaction.

- (b) Outline a procedure the student could follow to investigate the impact of temperature changes on this reaction. **2**

Question 23 (3 marks)

Outline the processes used by Aboriginal and Torres Strait Islander Peoples to remove toxic chemicals from food through an example.

3

This image shows a blank sheet of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the paper.

Question 24 (4 marks)

Draw models to show the difference between the following scenarios.

4

Strong Acid	Weak Acid

Concentrated strong Acid	Dilute strong Acid

Question 25 (4 marks)

Carbon monoxide reacts with chlorine to produce carbon oxychloride as shown by the following equation.



A 10 L reaction vessel initially contained 2.0 moles carbon monoxide and 5.0 moles chlorine gas at 200°C. When the reaction reached equilibrium, there was 1.0 mole of carbon monoxide remaining.

- (a) Calculate the equilibrium concentration of carbon monoxide, chlorine and carbon oxychloride. **2**

- (b) Write the equilibrium expression for this reaction then calculate the value for K_{eq} . Show relevant working. **2**

Question 26 (3 marks)

100.0 mL of a standard solution of sodium hydroxide (NaOH) was added to 100.0 mL of a 1.20 mol L⁻¹ solution of sulfuric acid in a solution calorimeter. The temperature of the solution rose from 18.6°C to 34.4°C.

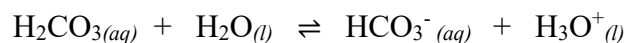
Determine the heat of neutralisation for this reaction. (Assume the density of the solution is 1.00 g mL^{-1}). Show all working.

3

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 27 (4 marks)

An equimolar solution of carbonic acid and hydrogen carbonate ions has reached equilibrium with a pH of 6.4.



- (a) What would happen if a small volume of dilute acid or base was added to this solution? **2**

- (b) Identify a buffer in a natural system and describe its importance in the identified system. **2**

Question 28 (4 marks)

Discuss how theories about acids and bases have developed over time as new observations were made and existing ideas were shown to be too limited.

4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 29 (7 marks)

A student was asked to determine the mass of CaCO_3 present in a 2.25 g sample of chalk. The chalk reacted with 100 mL of a 1.00 M HCl solution. After the bubbling stopped, excess HCl remained. The excess HCl was titrated with 1.0 M NaOH. The average titre was 17.15 mL.

- (a) Calculate the number of moles of sodium hydroxide which reacted with the excess HCl. 1

- (b) Determine the number of moles of HCl added to the chalk sample and use this to calculate the actual number of moles of calcium carbonate present in the sample. 4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (c) Calculate the mass of calcium carbonate present in the 2.25 g chalk sample. 2

[illegible]

Question 30 (3 marks)

Acetic acid is placed in two beakers. In the first beaker, an equal volume of water is added. In the second beaker, sodium hydroxide solution is added.

- (a) Write a balanced equation for the reaction between acetic acid and sodium hydroxide. **1**

.....

- (b) Account for the difference between the reactions taking place in the two beakers. **2**

.....

.....

.....

.....

Question 31 (3 marks)

The intermolecular bonding that forms in alkanes, alcohols and carboxylic acids determines physical properties of the substances.

- (a) Identify the type of intermolecular forces that form between alkanes. **1**

.....

.....

- (b) Compare the strength of the intermolecular bonding of alkanes to alcohols with the help of a diagram. **2**

.....

.....

.....

.....

.....

Question 32 (4 marks)

During your course of study, you carried out a first-hand investigation to compare the enthalpy of combustion of alcohols.

- (a) Identify a risk present in this investigation and outline the safety precaution required to minimise this risk. 1

- (b) A student recorded a temperature increase of 9.5°C when 1.4 g of methanol was combusted to heat 200 mL of water.

Given that the theoretical enthalpy of combustion of methanol is 726 kJ mol^{-1} , calculate the efficiency of the apparatus at absorbing the heat released during the combustion.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 33 (4 marks)

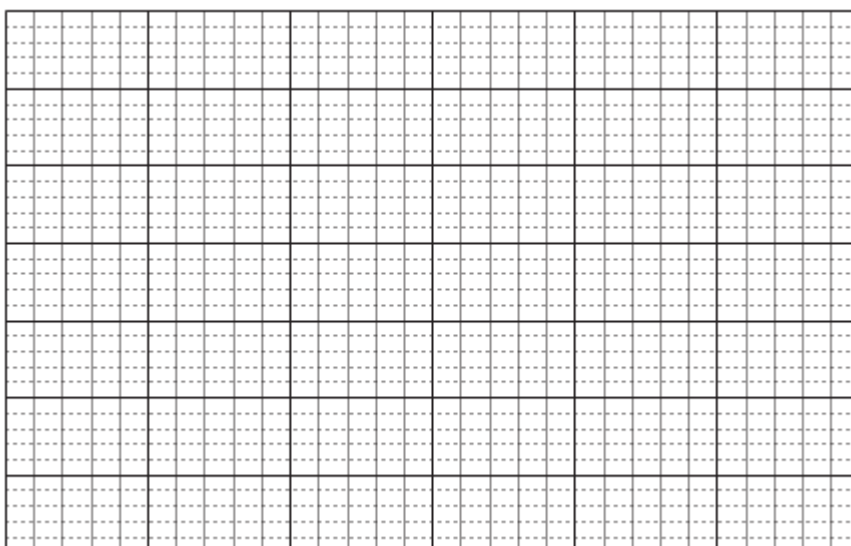
- (a) A volatile halogenated hydrocarbon needs to be disposed of safely. Describe the necessary precautions to safely handle and dispose of the hydrocarbon. 2

- (b) Hydrocarbons obtained from the Earth can be used to make plastic bags. Examine the environmental impact of sourcing and using hydrocarbons for this purpose.

[illegible]

The information in the table shows how the solubility of lead chloride is affected by temperature.

Using a graph, calculate the solubility product (K_{sp}) of the dissolution of lead chloride at 50°C. Include a fully labelled graph and a relevant chemical equation in your answer.

[illegible]

Question 35 (3 marks)

Calculate the pH of a 0.10 M acetic acid solution given $K_a = 1.8 \times 10^{-5}$. Show all working.

3

Question 36 (4 marks)

72.1 g of glucose is fermented anaerobically.

(a) Identify the organism used to ferment the glucose.

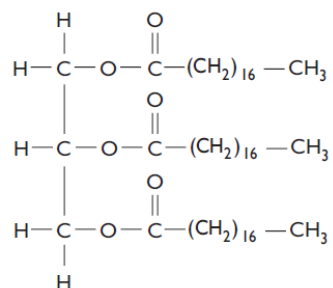
1

(b) Calculate the volume of carbon dioxide produced at 25°C and 100 kPa, given that one mole of gas occupies 24.79 L at these conditions.

3

Question 37 (5 marks)

The structure of a triglyceride present in naturally occurring fat is:



This triglyceride was hydrolysed by heating with concentrated sodium hydroxide.

(a) Write the equation of saponification of this triglyceride. 2

(b) Outline the method used for the saponification process in the laboratory. 2

.....

.....

.....

.....

.....

(c) Draw a simple schematic structure of a soap molecule. Label the hydrophobic and hydrophilic ends of this molecule. 1

Question 38 (7 marks)

The properties of three compounds, **X**, **Y** and **Z** are given in the table.

<i>Compound</i>	<i>Carboxylic acid</i>	<i>Primary Amine</i>	<i>Amide</i>
<i>Example</i>	CH ₃ COOH	CH ₃ CH ₂ CH ₂ NH ₂	CH ₃ CONH ₂
<i>Name</i>	X	Y	Z
<i>Molecular weight</i> (g mol ⁻¹)	60.052	59.112	59.068
<i>Boiling Point</i> (°C)	118	48	221
pK _a	4.756	10.71	0.63

- (a) Name the three compounds shown:

3

X _____
Y _____
Z _____

- (b) Write TWO equations to compare the reactions which occur when:

2

- (i) compound **X** is added to water
(ii) compound **Y** is added to water

.....
.....
.....
.....
.....

- (c) Explain the variation in boiling points between the carboxylic acid and the amine or the amide given that all three compounds have very similar molecular weights.

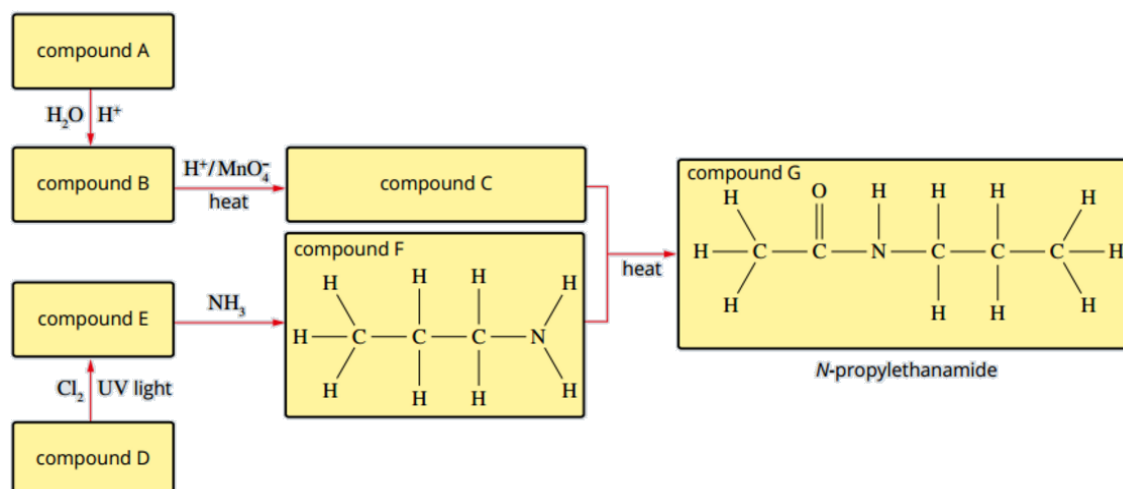
2

.....
.....
.....
.....
.....
.....

Question 39 (5 marks)

The reaction pathway below leads to the organic compound G (N-propylethanamide).

5



Draw and justify the structures of compounds A to E

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

2019 HSC TRIAL EXAMINATION

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

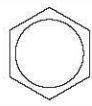
Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Infrared absorption data

Bond	Wavenumber/cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

¹³C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -C - C- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ R - C - Cl \text{ or } Br \\ \end{array}$	10–70
$\begin{array}{c} \\ R - C - C - \\ \quad \\ O \end{array}$	20–50
$\begin{array}{c} \\ R - C - N \\ \quad \diagup \end{array}$	25–60
$\begin{array}{c} \\ -C - O - \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \diagup \quad \diagdown \\ C = C \\ \diagdown \quad \diagup \end{array}$	90–150
R — C ≡ N	110–125
	110–160
$\begin{array}{c} R - C - \\ \\ O \end{array}$ esters or acids	160–185
$\begin{array}{c} R - C - \\ \\ O \end{array}$ aldehydes or ketones	190–220

UV absorption*(This is not a definitive list and is approximate.)*

Chromophore	λ _{max} (nm)
C—H	122
C—C	135
C=C	162

Chromophore	λ _{max} (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen																	2 He 4.003 Helium
3 Li 6.941 Lithium	4 Be 9.012 Beryllium															9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium															17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57-71 Lanthanoids	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89-103 Actinoids	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessee	118 Og Oganesson

KEY

Atomic Number	79
Symbol	Au
Standard Atomic Weight	197.0
Name	Gold

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
--------------------------------	-----------------------------	-----------------------------------	--------------------------------	------------------------	-------------------------------	-------------------------------	---------------------------------	------------------------------	---------------------------------	------------------------------	-----------------------------	------------------------------	--------------------------------	-------------------------------

Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
----------------------	------------------------------	-----------------------------------	-----------------------------	-----------------------	-----------------------	-----------------------	--------------------	-----------------------	-------------------------	-------------------------	----------------------	--------------------------	-----------------------	-------------------------

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

STUDENT ID: _____

2019 Chemistry HSC Trial Examination
Section I –Multiple Choice Answer Sheet

20 marks**Attempt Questions 1 –20****Allow about 35 minutes for this section**

Select the alternative A, B, C, or D that best answers the question. Fill in the response circle completely.

1	A ○	B ○	C ○	D ○
2	A ○	B ○	C ○	D ○
3	A ○	B ○	C ○	D ○
4	A ○	B ○	C ○	D ○
5	A ○	B ○	C ○	D ○
6	A ○	B ○	C ○	D ○
7	A ○	B ○	C ○	D ○
8	A ○	B ○	C ○	D ○
9	A ○	B ○	C ○	D ○
10	A ○	B ○	C ○	D ○
11	A ○	B ○	C ○	D ○
12	A ○	B ○	C ○	D ○
13	A ○	B ○	C ○	D ○
14	A ○	B ○	C ○	D ○
15	A ○	B ○	C ○	D ○
16	A ○	B ○	C ○	D ○
17	A ○	B ○	C ○	D ○
18	A ○	B ○	C ○	D ○
19	A ○	B ○	C ○	D ○
20	A ○	B ○	C ○	D ○

STUDENT ID: _____

MARKING GUIDE

2019 Chemistry HSC Trial Examination
Section I – Multiple Choice Answer Sheet

20 marks**Attempt Questions 1 –20****Allow about 35 minutes for this section**

Select the alternative A, B, C, or D that best answers the question. Fill in the response circle completely.

1	A ☐	B ☐	C ☐	D ☒
2	A ☐	B ☐	C ☐	D ☒
3	A ☐	B ☒	C ☐	D ☐
4	A ☒	B ☐	C ☐	D ☐
5	A ☒	B ☐	C ☐	D ☐
6	A ☐	B ☒	C ☒	D ☒
7	A ☐	B ☐	C ☒	D ☐
8	A ☐	B ☐	C ☐	D ☒
9	A ☐	B ☐	C ☒	D ☐
10	A ☐	B ☐	C ☐	D ☒
11	A ☒	B ☐	C ☐	D ☐
12	A ☐	B ☒	C ☐	D ☐
13	A ☐	B ☒	C ☐	D ☒
14	A ☐	B ☒	C ☐	D ☐
15	A ☐	B ☒	C ☐	D ☐
16	A ☐	B ☐	C ☒	D ☐
17	A ☒	B ☐	C ☐	D ☐
18	A ☐	B ☒	C ☐	D ☐
19	A ☒	B ☐	C ☐	D ☐
20	A ☐	B ☐	C ☒	D ☐



1. The first part of the document is a list of names and addresses of the members of the committee.

2. The second part of the document is a list of names and addresses of the members of the committee.

3. The third part of the document is a list of names and addresses of the members of the committee.

4. The fourth part of the document is a list of names and addresses of the members of the committee.

5. The fifth part of the document is a list of names and addresses of the members of the committee.

6. The sixth part of the document is a list of names and addresses of the members of the committee.

7. The seventh part of the document is a list of names and addresses of the members of the committee.

8. The eighth part of the document is a list of names and addresses of the members of the committee.

NEW MASTER

Killara HIGH SCHOOL

2019

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Remove the multiple-choice answer sheet at the back of Section I
- NESA approved calculators may be used
- A formula sheet, data sheet and Periodic Table are provided at the back of this paper
- For questions in Section II, show all relevant working in questions involving calculations

Total marks:
100

Section I — 20 marks (pages 2-8)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II — 80 marks (pages 9-25)

- Attempt Questions 21–39
- Allow about 2 hour and 25 minutes for this section

Section I

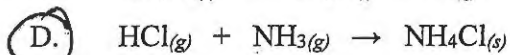
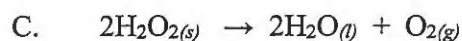
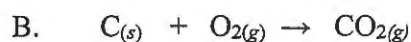
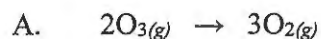
20 marks

Attempt Questions 1–20

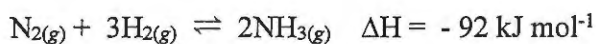
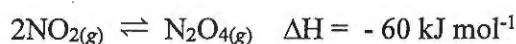
Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20

1 Which equation would represent the greatest decrease in entropy?



2 Consider the two reactions at equilibrium shown below:



Which action will drive BOTH reactions to form more product?

A. Reducing the overall pressure

B. Increasing the reaction temperature

C. Increasing the volume of the reaction vessel

☒ D. Removing the product from the reaction vessel3 Under what reaction conditions will the value of an equilibrium constant (K_{eq}) be changed?

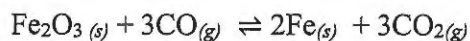
A. Addition of a catalyst

☒ B. Increased temperature

C. Increased concentration of reactants

D. Increased concentration of reactants and products

- 4 Which alternative most correctly represents the equilibrium expression for the following reaction?



- A. $\frac{[\text{CO}_2]^3}{[\text{CO}]^3}$
- B. $\frac{[\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3][\text{CO}]^3}$
- C. $\frac{[\text{Fe}_2\text{O}_3][\text{CO}]^3}{[\text{Fe}]^2[\text{CO}]^3}$
- D. $\frac{[\text{Fe}]^2[\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3][\text{CO}]^3}$ ✓

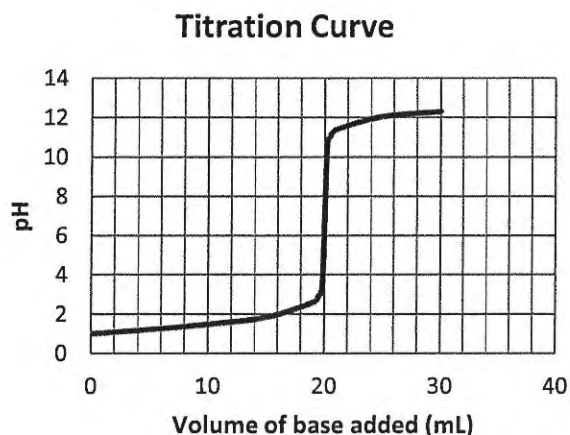
- 5 Which of the following reactions at equilibrium is closest to completion?

- A. $2\text{O}_3(g) \rightleftharpoons 3\text{O}_2(g) \quad K_{eq} = 2.0 \times 10^{57}$ ✓
- B. $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g) \quad K_{eq} = 4.3$
- C. $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g) \quad K_{eq} = 9.0$
- D. $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq) \quad K_{eq} = 1.8 \times 10^{-5}$

- 6 Which of the following is a Brønsted–Lowry reaction?

- A. $\text{CO}_2 + \text{OH}^- \rightleftharpoons \text{HCO}_3^-$
- B. $\text{NH}_4^+ + \text{NH}_2^- \rightleftharpoons 2\text{NH}_3$
- C. $\text{HClO}_4 + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{ClO}_4^-$
- D. $\text{CH}_3\text{CH}_2\text{O}^- + \text{CH}_3\text{NH}_3^+ \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{NH}_2$ ✓

Use the following information for Questions 7 and 8.



- 7 Which statement about this titration is correct?
- A. The acid was placed in the burette and the equivalence point was at pH 7.
 - B. The base was placed in the burette and the equivalence point was at pH 12.
 - ☒ C. The acid was placed in the conical flask and the equivalence point was at pH 7. ✓
 - D. The acid was placed in the conical flask and the equivalence point was at pH 12.
- 8 Which pair of solutions would be most likely to produce this titration curve?
- A. Acetic acid and ammonia solution
 - B. Citric acid and sodium hydroxide
 - C. Sulfuric acid and ammonia solution
 - ☒ D. Hydrochloric acid and sodium hydroxide ✓
- 9 What is the concentration of H_3O^+ in a solution with a pH of 2.38?
- A. 0.38 mol L^{-1}
 - B. 11.62 mol L^{-1}
 - ☒ C. $4.17 \times 10^{-3} \text{ mol L}^{-1}$ ✓
 - D. $2.40 \times 10^{-12} \text{ mol L}^{-1}$

10 Which alternative contains only amphoteric substances?

- A. SiO_2 , SO_2 , SnO
- B. H_2CO_3 , HCO_3^- , CO_3^{2-}
- C. HS^- , H_3PO_4 , HPO_4^{2-}
- ☒ D. HCO_3^- , HSO_4^- , H_2O

11 25.00 mL of 0.02 M potassium hydroxide solution in a conical flask is neutralised by the gradual addition of 33.20 mL of nitric acid.

What is the concentration of the nitric acid?

- ☒ A. 0.015 mol L^{-1}
- B. $0.0265 \text{ mol L}^{-1}$
- C. $5 \times 10^{-4} \text{ mol L}^{-1}$
- D. $1.66 \times 10^{-5} \text{ mol L}^{-1}$

12 The formula of potassium hydrogen iodate is:

- A. $\text{HK}(\text{IO}_3)_2$
- ☒ B. $\text{KH}(\text{IO}_3)_2$
- C. $\text{H}_2\text{K}(\text{IO}_3)_2$
- D. $\text{K}_2\text{H}(\text{IO}_3)_2$

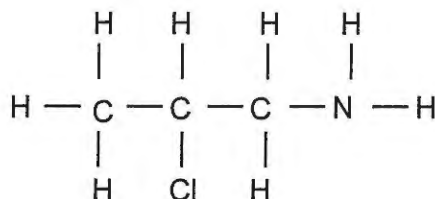
13 Which of the following is the conjugate base of the dihydrogen phosphate ion (H_2PO_4^-)?

- A. OH^-
- B. PO_4^{3-}
- C. H_3PO_4
- ☒ D. HPO_4^{2-}

14 Which condensed formula represents a chain isomer for $C_4H_8F_2$?

- A. $(CH_3)_2CHFCHF$
- ☒ B. $CH_3CHFCHFCH_3$
- C. $CH_3CH_2CHFCHF$
- D. $CH_2FCHCHFCH_3$

15 Consider the following structure.

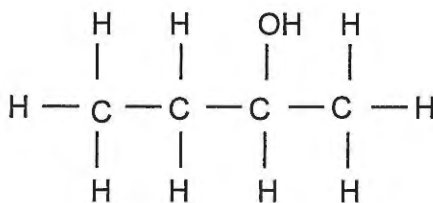


What is the IUPAC name for this molecule?

- A. 2-chloropropanamide
- ☒ B. 2-chloropropan-1-amine
- C. 2-chloropropan-3-amine
- D. 3-amino-2-chloropropane



16 Acidified potassium permanganate solution is a powerful oxidising agent. It is added to a substance with the following structural formula;



What is the name of the organic product of this reaction?

- A. Butan-2-ol
- B. Butanal
- ☒ C. Butan-2-one
- D. Butanoic acid



- 17 Three separate colourless solutions each contain one cation, Na^+ , Pb^{2+} or Ca^{2+} . Which of the following would be an appropriate reagent to unambiguously identify the solution containing Pb^{2+} ?

- A. KI
 B. K_2CO_3
 C. K_3PO_4
 D. AgNO_3



- 18 A colourless organic liquid compound is being tested for the presence of functional groups. 2 mL of the liquid is poured into three separate test tubes. A reagent is added to each test tube then shaken.

Test	Reagent added	Observation made after shaking
1	2 drops of bromine water	Yellowish-brown coloured solution
2	3 drops of acidified potassium dichromate	Yellowish-orange coloured solution
3	1 g of sodium hydrogen carbonate	Colourless solution, small bubbles of colourless gas

What functional group(s) may be present in this compound?

- A. Double bonds
 B. Carboxylic acid
 C. Double bonds and hydroxyl group
 D. Hydroxyl group bonded to a secondary carbon

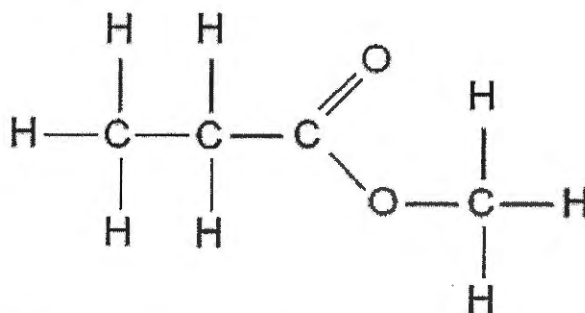


- 19 Which of the following is a correct statement about primary and tertiary alcohols?

- A. Primary alcohols can be oxidised to aldehydes while tertiary alcohols cannot be oxidised.
 B. Primary alcohols can be oxidised to aldehydes while tertiary alcohols are oxidised to ketones.
 C. Primary alcohols have one hydroxyl functional group while tertiary alcohols have three hydroxyl functional groups.
 D. Primary alcohols always have a hydroxyl functional group on the terminal carbon of the parent chain while tertiary alcohols have their functional group on the third carbon of the parent chain.



20 The formula of an ester is shown below.



Which alkanol and alkanoic acid were used to make this ester?

- A. Propanol and methanoic acid
- B. Methanol and ethanoic acid
- ☒ C. Methanol and propanoic acid
- D. Ethanol and methanoic acid

✓

2019**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Chemistry

Section II

Answer Booklet

80 marks**Attempt Questions 21–39****Allow about 2 hours and 25 minutes for this section**

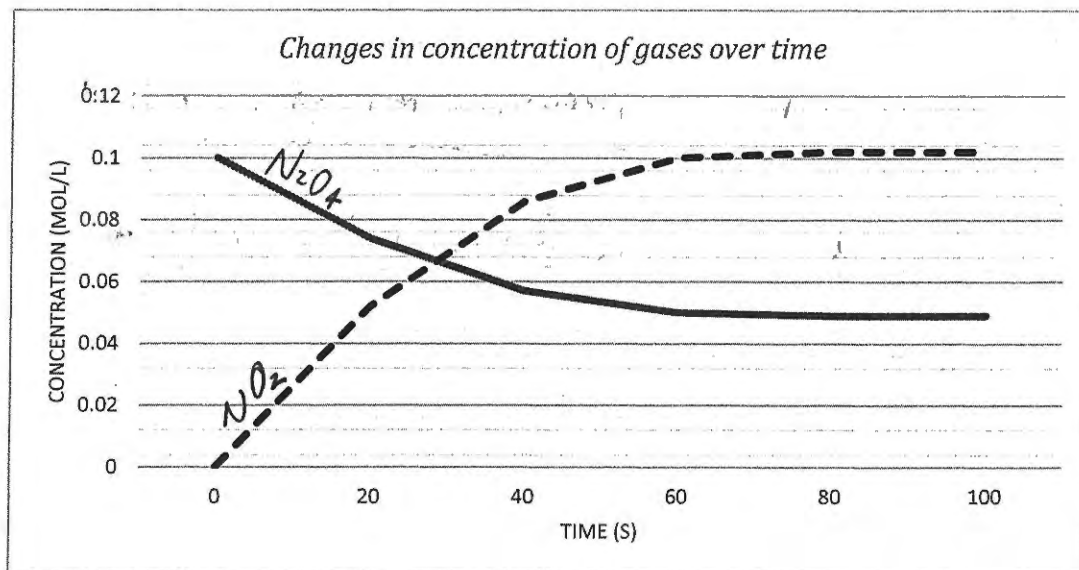
Instructions

- Write your Student ID above
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in question involving calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

Question 21 (4 marks)

A sealed glass flask originally contained colourless dinitrogen tetroxide gas with a concentration of 0.100 mol L^{-1} at a temperature of 100°C . The dinitrogen tetroxide decomposed to form a dark brown gas.



- (a) Describe the macroscopic appearance of the contents of the flask after 80 seconds.

light brown gas //
or brown //
or yellow (not dark brown)

1

1
(no halves)

- (b) Describe what happened at the molecular level to the contents of the flask after 80 seconds.

molecules of N_2O_4 decompose into NO_2 molecules
at the same rate that NO_2 molecules (collide) to
form N_2O_4 molecules
(if they just said reached equilibrium/similar $\Rightarrow \frac{1}{2}$)

1

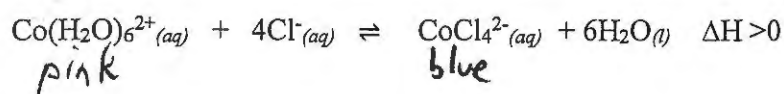
- (c) Explain the changes in reaction rate from 0 to 80 seconds.

2

marked hard $\rightarrow$ described changes in reactants + products $\Rightarrow 1$
explained in terms of particles colliding $\Rightarrow 2$ (v. few students did this)
eg. Initially $[\text{N}_2\text{O}_4]$ molecules (reactants) is high, but as reactants molecules get
used up, the forward reaction slows down because there are
fewer (successful) collisions as conc² decreases. The rate of the reverse
reaction is slow initially as there are few NO_2 molecules. As more product
is formed, the reverse reaction speeds up until a dynamic equilibrium
is reached

Question 22 (4 marks)

A student was provided with a beaker containing a violet coloured solution of cobalt (II) chloride and was asked to investigate the following equilibrium reaction:



Note: pink + blue
↓
violet

- (a) 5 mL of cobalt (II) chloride solution was poured into a test tube with 5 drops of 4 M HCl. 2

1 mark
Use Le Chatelier's principle to predict the impact on this reaction. 1 mark → ½ blue
→ ½ to the right due to Cl^{-} ↑

eg. The addition of HCl raises the concⁿ of Cl^{-} causing the equil^m to shift to the right to reduce the concⁿ of added Cl^{-} (as predicted by Le Chatelier Principle)

This solution changes from violet to blue (as more CoCl_4^{2-} forms.)
(Note: can't justify a wrong answer eg. to the left)

- (b) Outline a procedure the student could follow to investigate the impact of temperature changes on this reaction. 2

a range of temperatures used / control + hot + cold ⇒ 1 mark

tubes/beakers ½ record colour changes ½

Question 23 (3 marks)

Outline the processes used by Aboriginal and Torres Strait Islander Peoples to remove toxic chemicals from food through an example.

3

named example eg. Cycad $\Rightarrow$ ① (seed $\frac{1}{2}$ not $\frac{1}{2}$ unless named)

roasting/crushing/boiling/slicing/pounding ① $\frac{1}{2}$

leaching / toxin removed by washing
removed by running water ①

toxin is soluble/dissolves in water ① $\frac{1}{2}$

Question 24 (4 marks)

Draw models to show the difference between the following scenarios.

4

Strong Acid	Weak Acid
Any model that discriminated between each type of acid Teach but; - need a key/info/labels to identify particles (so teacher does not have to interpret their diagram)	
Concentrated strong Acid	Dilute strong Acid

Question 25 (4 marks)

Carbon monoxide reacts with chlorine to produce carbon oxychloride as shown by the following equation.



A 10 L reaction vessel initially contained 2.0 moles carbon monoxide and 5.0 moles chlorine gas at 200°C. When the reaction reached equilibrium, there was 1.0 mole of carbon monoxide remaining.

- (a) Calculate the equilibrium concentration of carbon monoxide, chlorine and carbon oxychloride. 2

Species	Initial Conc ⁿ	Change	Equil ^m Conc ⁿ
CO	0.2	-0.1	0.1 (given)
Cl ₂	0.5	-0.1	0.4
COCl ₂	0	∴ +0.1	0.1

∴ CO 0.1 mol L⁻¹ / mark for correct calculation
 Cl₂ 0.4 mol L⁻¹
 COCl₂ 0.1 mol L⁻¹ / mark for correct answer

Note: accepted M as units but should not have

- (b) Write the equilibrium expression for this reaction then calculate the value for K_{eq}. 2
 Show relevant working.

$$K = \frac{[\text{COCl}_2]}{[\text{CO}][\text{Cl}_2]}$$

1 mark

$$= \frac{0.1}{0.1 \times 0.4}$$

$$= 2.5 \quad \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \text{ for 2 sig. figs}$$

Question 26 (3 marks)

100.0 mL of a standard solution of sodium hydroxide (NaOH) was added to 100.0 mL of a 1.20 mol L⁻¹ solution of sulfuric acid in a solution calorimeter. The temperature of the solution rose from 18.6°C to 34.4°C.

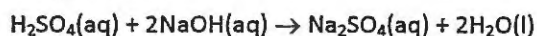
Determine the heat of neutralisation for this reaction. (Assume the density of the solution is 1.00 g mL^{-1}). Show all working.

1 mark: Number of moles of H_2SO_4 and H_2O

1 mark: $q=mc\Delta T$

1 mark: $\Delta H = \frac{q}{n}$

Sample answer:



$$n(\text{H}_2\text{SO}_4) = c \times V = 0.1 \times 1.2 = 0.120 \text{ mol}$$

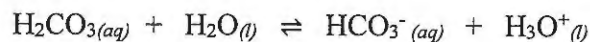
$$q = m \times c \times \Delta T = 200 \times 4.184 \times 15.8 = 13\,200 \text{ J} = 13.2 \text{ kJ}$$

$$n(\text{H}_2\text{O}) = 2n(\text{H}_2\text{SO}_4) = 2 \times 0.12 = 0.24 \text{ mol}$$

$$\Delta H = \frac{13.2}{0.24} = 55.0 \text{ kJ mol}^{-1}$$

Question 27 (4 marks)

An equimolar solution of carbonic acid and hydrogen carbonate ions has reached equilibrium with a pH of 6.4.



- (a) What would happen if a small volume of dilute acid or base was added to this solution? 2

1 mark: Recognition that this is a buffer system

1 mark: Addition of acid or base will shift the equilibrium (1/2) so that the pH remains the same (1/2)

Note: just explaining shifting according to LCP: ½ mark

Sample answer:

An equimolar solution of $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ is a buffer. This is a solution consisting of a weak acid and its conjugate base.

If a small amount of acid such as HCl is added to this buffer then the equilibrium will shift to the left as the increase in hydrogen ions will cause the reaction to shift in such a way as to counteract the change and the pH will remain constant.

If a small amount of base such as NaOH is added to this buffer then the equilibrium will shift to the right as the increase in hydroxide will cause the reaction to shift in such a way as to counteract the change and the pH will remain constant.

- (b) Identify a buffer in a natural system and describe its importance in the identified system. 2

1 mark: Correct buffer system with chemical equation (such as carbonate in ocean, haemoglobin in blood, hydrogen carbonate in river/lake)

1 mark: Explanation how this buffer system remains the pH

Question 28 (4 marks)

Discuss how theories about acids and bases have developed over time as new observations were made and existing ideas were shown to be too limited.

4

Criteria	Marks
• Demonstrates thorough knowledge and understanding of changing theories about acids and bases • Identifies observations which lead to multiple modifications to theories • Names numerous chemists who proposed theories	4
• Demonstrates sound knowledge and understanding of changing theories about acids and bases • Identifies observations which lead to multiple to theories • Names numerous chemists who proposed theories	3
• Demonstrates some knowledge and understanding of changing theories about acids and bases • Identifies observations which lead to modifications to theories • Names chemists who proposed theories	2
• Provides some information about a theory of acids and bases • Names a chemist who proposed theory	1

Sample answer:

An early definition of acids was made by Lavoisier who said that acids were compounds which contained the element oxygen. When acids were discovered such as hydrochloric acid that did not contain oxygen the definition was adjusted by Davy to acids were compounds that contained the element hydrogen. It was then observed that both acids and bases conducted electricity.

Arrhenius improved on Davy's ideas by theorising that acids were compounds that produced hydrogen ions in water and bases produced hydroxide ions however it was shown that substances such as ammonia could behave like a base but did not contain hydroxide ions. The concept of acids and bases was further developed by two chemists, Brønsted and Lowry, who said that acids are chemical species that transfer a proton to another species while bases were substances that received a proton. They can explain why ammonia is a base (equation). However, Brønsted and Lowry could not explain why compounds such as BF_3 and AlCl_3 are acid but have no protons nor they could not explain the reactions of oxides that take place without solvents. Lewis then proposed a new definition of acids and bases, acids are electron pair acceptors while bases are electron pair donor. This new definition enable scientists to explain why BF_3 and AlCl_3 are acid (equation)

Question 29 (7 marks)

A student was asked to determine the mass of CaCO_3 present in a 2.25 g sample of chalk. The chalk reacted with 100 mL of a 1.00 M HCl solution. After the bubbling stopped, excess HCl remained. The excess HCl was titrated with 1.0 M NaOH. The average titre was 17.15 mL.

- (a) Calculate the number of moles of sodium hydroxide which reacted with the excess HCl. 1

1 mark: number of moles ($n = c V = 1.0 \times 17.15 \times 10^{-3} = 1.715 \times 10^{-2} \text{ mol}$)

- (b) Determine the number of moles of HCl added to the chalk sample and use this to calculate the actual number of moles of calcium carbonate present in the sample. 4

1 mark: number of mole of HCl added to the chalk (0.10 mol)

1 mark: number of mole HCl reacted with NaOH ($1.715 \times 10^{-2} \text{ mol}$)

1 mark: number of mole HCl reacted with chalk ($8.285 \times 10^{-2} \text{ mol}$)

1 mark: number of mole CaCO_3 in the chalk sample ($4.1425 \times 10^{-2} \text{ mol}$)

Note: need to show logical working out

- (c) Calculate the mass of calcium carbonate present in the 2.25 g chalk sample. 2

1 mark: molar mass of CaCO_3 ($100.09 \text{ g mol}^{-1}$)

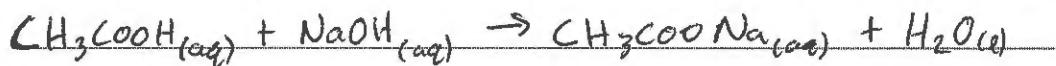
1 mark: mass of CaCO_3 in the chalk 4.14 g

-1/2 mark off for no unit

Question 30 (3 marks)

Acetic acid is placed in two beakers. In the first beaker, an equal volume of water is added. In the second beaker, sodium hydroxide solution is added.

- (a) Write a balanced equation for the reaction between acetic acid and sodium hydroxide. 1



- (b) Account for the difference between the reactions taking place in the two beakers. 2

Beaker 1 - ionised in presence of water

Beaker 2 - neutralisation reaction

Neutralisation AND ionisation ... (2)

neutralisation AND dilution/dissociation ... (1)

Question 31 (3 marks)

The intermolecular bonding that forms in alkanes, alcohols and carboxylic acids determines physical properties of the substances.

- (a) Identify the type of intermolecular forces that form between alkanes. 1

Dispersion forces ... (1)

- (b) Compare the strength of the intermolecular bonding of alkanes to alcohols with the help of a diagram. 2

Both alkanes and alcohols exhibit weak dispersion forces between molecules, however, the presence of the hydroxyl (-OH) group in alcohols leads to hydrogen bonding between alcohols which is a stronger interaction. The diagram shows the difference between interactions of molecules.

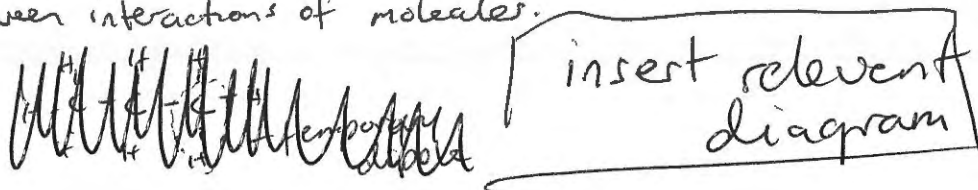


Diagram links to text AND relevant ... (2)

H-bonding AND dispersion AND comparison ... (1)

(-1/2
no mention
of
H-bond or
dispersion)

Question 32 (4 marks)

During your course of study, you carried out a first-hand investigation to compare the enthalpy of combustion of alcohols.

- (a) Identify a risk present in this investigation and outline the safety precaution required to minimise this risk. 1

relate risk to precaution AND specific to experiment ... (1)
 eg. volatile organics stored away from open flames.
 • anything reasonable.

* A general statement referring to safety glasses/hair tied back was not awarded marks

- (b) A student recorded a temperature increase of 9.5°C when 1.4 g of methanol was combusted to heat 200 mL of water. 3

Given that the theoretical enthalpy of combustion of methanol is 726 kJ mol^{-1} , calculate the efficiency of the apparatus at absorbing the heat released during the combustion.

$$\Delta T = 9.5\text{ K} \quad m = 0.200\text{ kg} \quad c = 4.18 \times 10^3\text{ J kg}^{-1}\text{K}^{-1}$$

$$q = mc\Delta T$$

$$= 0.2 \times 4.18 \times 10^3 \times 9.5$$

$$= 7942\text{ J}$$

$$M(\text{methanol}) = 12.01 + 4 \times 1.008 + 16.00$$

$$= 32.042\text{ g mol}^{-1}$$

$$n(\text{methanol}) = \frac{m}{M} = \frac{1.4\text{ g}}{32.042\text{ g mol}^{-1}} = 4.369 \times 10^{-2}\text{ mol}$$

$$\Delta H_c = \frac{7942}{4.369 \times 10^{-2}} = 181769.69 = 182\text{ kJ mol}^{-1}$$

$$\therefore \text{efficiency} = \frac{182}{726} \times 100 = 25.07\% = 25\%$$

Read
the
question

CH_4
is
not
 CH_3OH

calculates efficiency (most compare kJ mol^{-1} with kJ mol^{-1}) ... (3)

correct sub for q and calculation of ... moles ... (2)

write $q = mc\Delta T$ or equivalent ... (1)

Question 33 (4 marks)

- (a) A volatile halogenated hydrocarbon needs to be disposed of safely. Describe the necessary precautions to safely handle and dispose of the hydrocarbon.

2

A volatile halogenated hydrocarbon should only be used in a well ventilated area or fume hood. It should be disposed in a halogenated organic waste container.

safely handle - specific AND ventilation/fume hood

dispose - organic waste container

- (b) Hydrocarbons obtained from the Earth can be used to make plastic bags. Examine the environmental impact of sourcing and using hydrocarbons for this purpose.

2

sourcing - link to environmental impact in specific terms

using (plastic bags) - link to environmental impact in specific terms

just saying pollution, etc. was too general

Sourcing hydrocarbons through fracking can lead to a contaminated water supply and affect the surrounding environment through the loss of species. Creating ~~single use~~ plastic bags that are usually single use is problematic for the environment as they take a long time to break down releasing microplastics and affecting animal life or causing deaths in marine life.

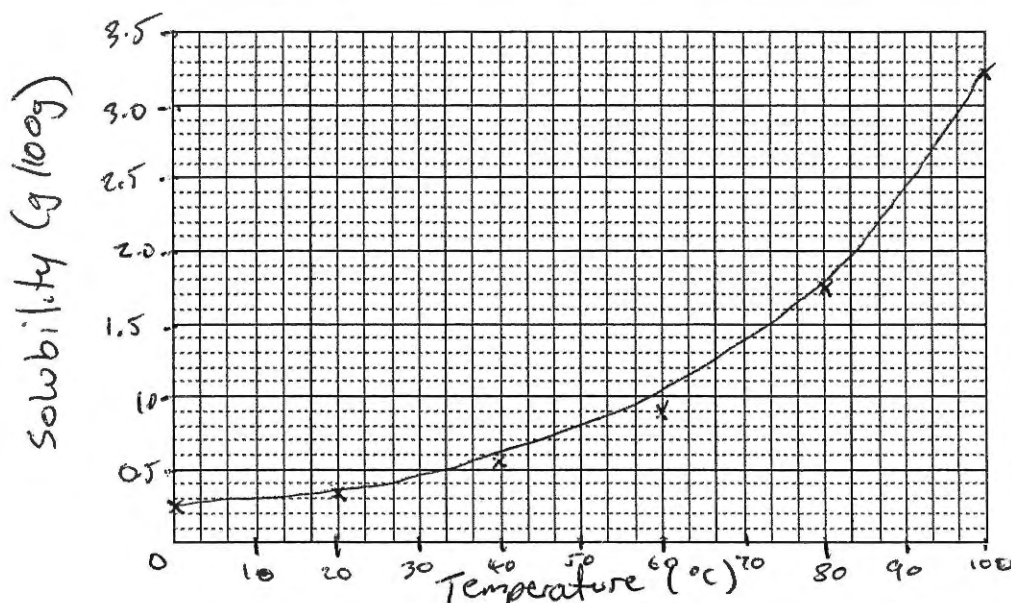
Question 34 (5 marks)

The information in the table shows how the solubility of lead chloride is affected by temperature.

5

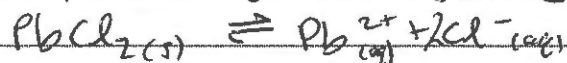
Temperature (°C)	Solubility (g/100 g water)
0	0.25
20	0.35
40	0.55
60	0.90
80	1.75
100	3.20

Using a graph, calculate the solubility product (K_{sp}) of the dissolution of lead chloride at 50°C. Include a fully labelled graph and a relevant chemical equation in your answer.



graph --- (1)
 Value from graph --- (1)
 equation --- (1)
 $n(\text{PbCl}_2)$
 OR
 $[\text{Pb}^{2+}]$ and $[\text{Cl}^-]$ --- (1)
 K_{sp} value --- (1)
 (from mol/L^{-1})

At 50°C, solubility = 0.8 g/100g



$$n(\text{PbCl}_2) = \frac{0.8}{207.2 + 35.45 \times 2} = 2.877 \times 10^{-3} \text{ mol}$$

$$\therefore \frac{2.877 \times 10^{-3} \text{ mol}}{0.1 \text{ L}} = 2.877 \times 10^{-2} \text{ mol/L}^{-1}$$

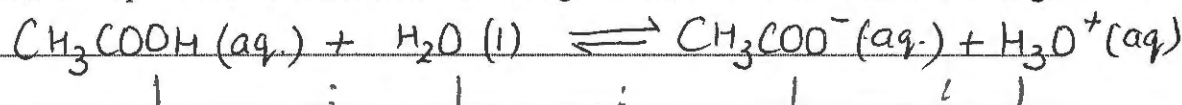
$$[\text{Pb}^{2+}] = 2.877 \times 10^{-2} \text{ mol/L}^{-1}$$

$$[\text{Cl}^-] = 2.877 \times 10^{-2} \times 2 = 5.753 \times 10^{-2} \text{ mol/L}^{-1}$$

$$K_{sp} = [\text{Pb}^{2+}][\text{Cl}^-]^2 = 9.523 \times 10^{-5}$$

Question 35 (3 marks)

Calculate the pH of a 0.10 M acetic acid solution given $K_a = 1.8 \times 10^{-5}$. Show all working. 3



$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}]}$$

$$1.8 \times 10^{-5} = \frac{[\text{H}_3\text{O}^+]^2}{0.10} = x^2 \quad (1:1 \text{ molar ratio})$$

$$x \text{ or } [\text{H}_3\text{O}^+] = \sqrt{1.8 \times 10^{-5} \times 0.10} = 0.00134164078 \text{ M}$$

$$\text{pH} = -\log_{10}[0.00134164078] = 2.872 = 2.9$$

Question 36 (4 marks)

72.1 g of glucose is fermented anaerobically.

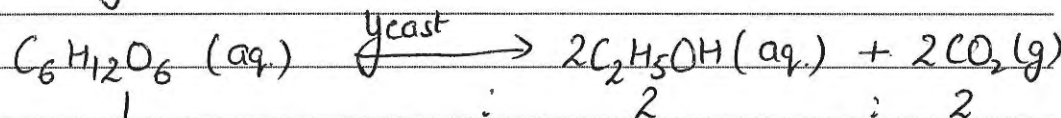
- (a) Identify the organism used to ferment the glucose. 1

Yeast

- 1 Mark

- (b) Calculate the volume of carbon dioxide produced at 25°C and 100 kPa, given that one mole of gas occupies 24.79 L at these conditions. 3

$$n_{\text{glucose}} = 72.1 \text{ g}$$



$$n_{\text{glucose}} = \frac{m}{MM} = \frac{72.1}{(12.01 \times 6 + 1.008 \times 12 + 16 \times 6)} = \frac{72.1}{180.16} = 0.4002 \text{ mols}$$

$$n_{\text{CO}_2} = 2 \times n_{\text{glucose}} \quad (2:1 \text{ molar ratio})$$

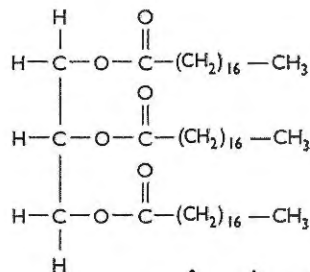
$$= 2 \times 0.4002 = 0.800417 \text{ mols.}$$

$$V_{\text{CO}_2} = n \times MV = 0.800417 \times 24.79$$

$$= 19.81 \text{ L}$$

Question 37 (5 marks)

The structure of a triglyceride present in naturally occurring fat is:



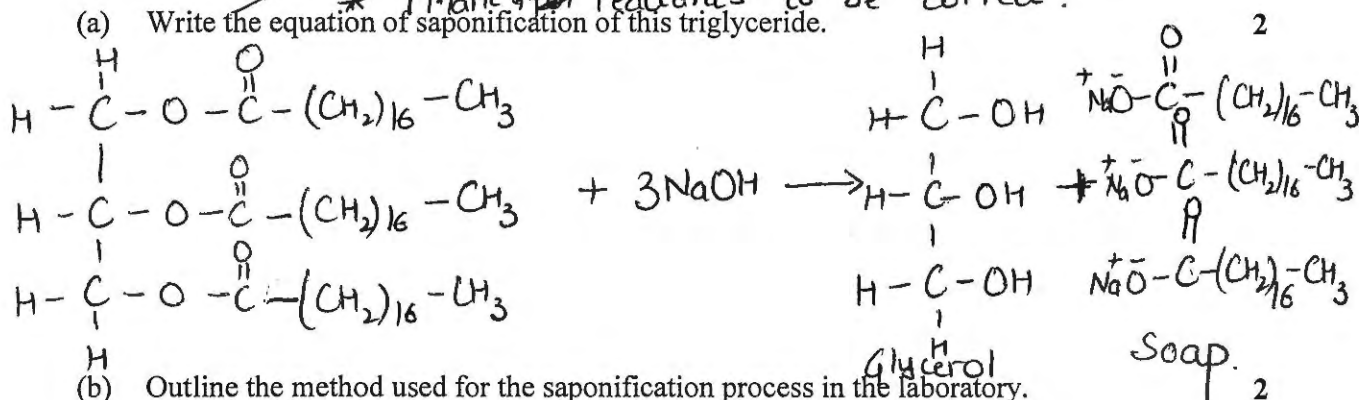
* Encourage students to write full structural formula eq. to avoid making mistakes.

* Condensed structural formula accepted if correct.

This triglyceride was hydrolysed by heating with concentrated sodium hydroxide.

* $\frac{1}{2}$ mark if triglyceride correct & NaOH (not 3NaOH)

(a) Write the equation of saponification of this triglyceride. * 1 Mark for reactants to be correct.



(b) Outline the method used for the saponification process in the laboratory.

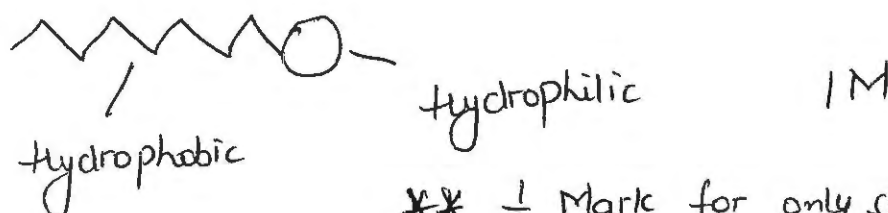
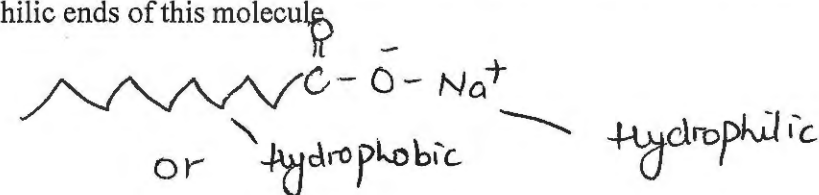
Veg oil / coconut oil + sodium hydroxide in a beaker $\rightarrow \frac{1}{2}$ M

Add ethanol-water mixture to dissolve NaOH $\rightarrow \frac{1}{2}$ M

Heat on a hot plate $\rightarrow \frac{1}{2}$ M

Salting out / Add salt / sat salt solution $\rightarrow \frac{1}{2}$ M

(c) Draw a simple schematic structure of a soap molecule. Label the hydrophobic and hydrophilic ends of this molecule



** $\frac{1}{2}$ Mark for only structure provided.



Question 38 (7 marks)

The properties of three compounds, **X**, **Y** and **Z** are given in the table.

Compound	Carboxylic acid	Primary Amine	Amide
Example	CH ₃ COOH	CH ₃ CH ₂ CH ₂ NH ₂	CH ₃ CONH ₂
Name	X	Y	Z
Molecular weight (g mol ⁻¹)	60.052	59.112	59.068
Boiling Point (°C)	118	48	221
pK _a	4.756	10.71	0.63

- (a) Name the three compounds shown:

3

X Ethanoic acid / Acetic acid - 1M

Y Propan-1-amine - 1M 1-Propanamine - ½M

Z Ethanamide - 1M

- (b) Write TWO equations to compare the reactions which occur when:

2

(i) compound **X** is added to water

(ii) compound **Y** is added to water



* no marks for '⇌' sign.

* equation 2 was written poorly across the board.

- (c) Explain the variation in boiling points between the carboxylic acid and the amine or the amide given that all three compounds have very similar molecular weights.

2

½M - relating energy needed to strength of H-bonding.

1M - identified all the structures have H-bonding.

½M - include degree of H-bonding / strongest in amide, followed by carboxylic acid & then amine.

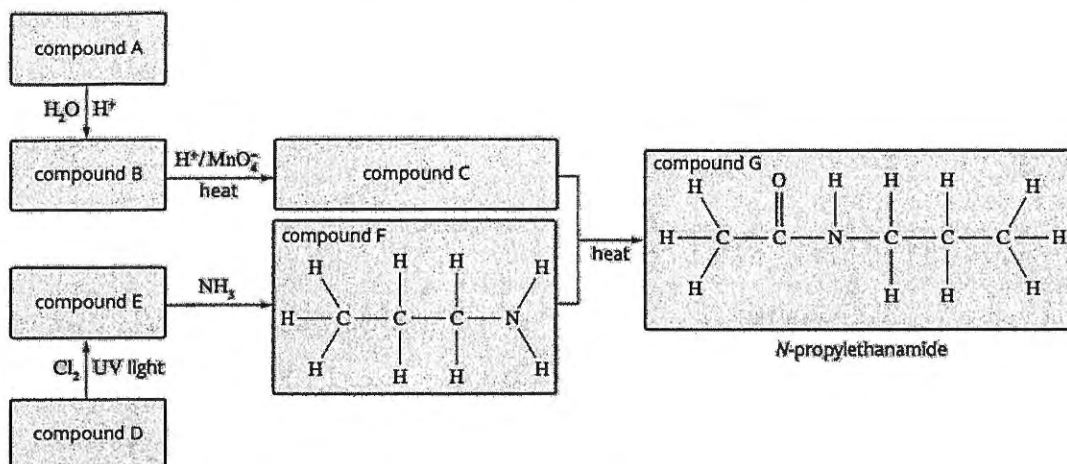
or reason using electronegativity difference to explain stronger intermolecular forces / dimer formation etc.

* Comparison was needed for only 2 compounds.

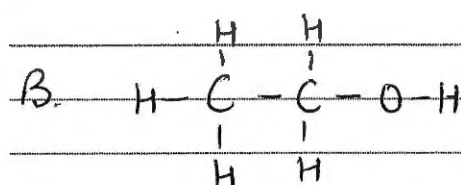
Question 39 (5 marks)

The reaction pathway below leads to the organic compound G (N-propylethanamide)

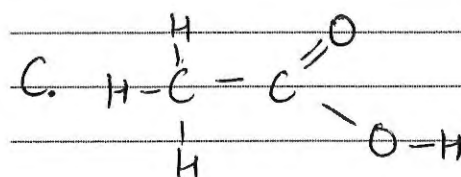
5



Draw and justify the structures of compounds A to E

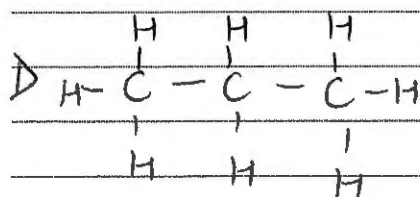


* 1 Mark for 1 Justification only.
(could be name of the reaction or explanation of it)



eg. Adding Cl_2 to an alkane requires UV light. so Compound C is propane.

or $\text{A} \rightarrow \text{B}$ is hydration etc or addition of water.



* Last 2 ticks - $\frac{1}{2}$ M each.

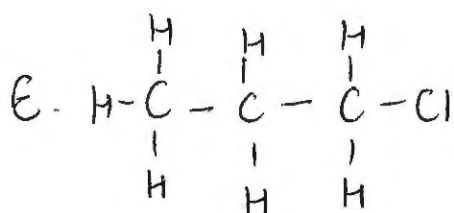
i.e if you got 1 structure

wrong you'll lose $\frac{1}{2}$ M

End of paper

& if you got 2 wrong (-1M)

* Rest of the ticks/correct answers - 1M each.



* This & was not answered very well across the board.